

Exercise for Sect. 2.6

Exercise

2.6.1

Let $X = \{x_1, \dots, x_r\}$ be a finite set, and let $\sigma : X \rightarrow X$ be a permutation of X . The orbit of x_j under σ is the set $\{\sigma^n(x_j)\}_{n \geq 0}$, and σ is called cyclic if there is an orbit of cardinality r .

1. For a cyclic permutation σ and any function $f : X \rightarrow \mathbb{R}$, prove that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} f(\sigma^j x) = \frac{1}{r} (f(x_1) + \dots + f(x_r)).$$

2. More generally, prove that for any permutation σ and function $f : X \rightarrow \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} f(\sigma^j x) = \frac{1}{p_x} (f(x) + f(\sigma(x)) + \dots + f(\sigma^{p_x-1}(x)))$$

where the orbit of x has cardinality p_x under σ .

Proof:

□

Exercise**2.6.2**

Minic the proof of finite Vitali covering lemma to prove that for any collection of intervals

$$I_1 = [a_1, a_1 + l(1) - 1], \dots, I_K = [a_K, a_K + l(K) - 1]$$

in $\mathbb{Z}$ there is a disjoint subcollection $I_{j(1)}, \dots, I_{j(K)}$ such that

$$I_1 \cup \dots \cup I_K \subseteq \bigcup_{m=1}^k [a_{j(m)} - l(j(m)), a_{j(m)} + 2l(j(M)) - 1].$$

Proof:

□

Exercise**2.6.3**

Let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system. Prove that, for any

$$f \in L^1_\mu,$$

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^{-n} x)$$

almost everywhere.

Proof:

□

Exercise

2.6.4

Fill in the details to prove the estimate in (2.28).

Proof:

□

Exercise

2.6.5

Formulate and prove a pointwise ergodic theorem for a measurable function $f \geq 0$ with $\int f \, d\mu = \infty$, under the assumption of ergodicity.

Proof:

□